Hybrid versus intrinsic Majorana nanowires, a self-consistent Hartree-Fock-Bogoliubov analysis

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We theoretically investigate hybrid and intrinsic Majorana nanowires accounting for self-consistent superconductive correlations (as opposed to an extrinsically fixed pairing) in a Hartree-Fock-Nambu mean-field theory. For this, we Hartree-Fock decouple a quartic Hamiltonian in a Nambu-spinor scheme accounting for the anomalous terms, extracting the mean-field self-energy as a spin-1/2 Hubbard term in the Wannier limit. We discuss beyond minimal models how the combined effect of spin-orbit coupling and Zeeman field tangled with the emergent self-consistent superconductivity triggers a trivial-topological phase transition, along with the role of finite length on the emergence of Majorana oscillations. We show that Majoranas do still emerge in the hybrid case, although under certain restrictions due to not accounting for disorder (ballistic regime), but most importantly that they emerge in a precarious balance in the intrinsic case, not being fully topological protected.

I. INTRODUCTION

1. What is a Majorana nanowire
2. The hybrid approach
3. The problems of the hybrid approach
4. The prospect of intrinsic nanowires
5. The importance of self-consistency
6. The goal of this work

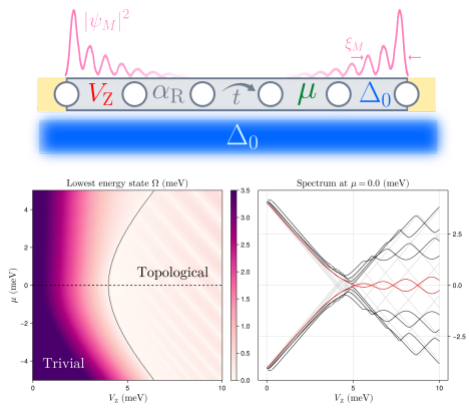

Figure 1. Minimal Majorana model

In Lutchyn's minimal model, having set a value for the uniform pairing Δ_0, one can induce a trivial-topological phase transitions by the tuning of the Zeeman field V_Z and chemical potential μ. This non-symmetry protected phase transition occurs exactly at the frontier of the critical value $V_Z^c = \sqrt{\mu^2 + \Delta_0^2}$ signaled by the closing and reopening of the band gap: for $V_Z < V_Z^c$ the band spectrum has a large gap due to while for $V_Z > V_Z^c$ it maintains a pairing minigap due to the combined affects of Zeeman and SOC with the emerging of topological protected Majorana zero energy modes.

We mainly focus the analysis on how the spectrum and superconductive correlations behaves as a function of Zeeman and chemical potential, more particularly the closing and reopening of the gap, with additional subtlety of self-consistence. For both hybrid and intrinsic models, we start by examining the cases of infinite length nanowires since these lay out much of the necessary groundwork without muddying it with finite length effects.

II. SELF-CONSISTENT SUPERCONDUCTIVITY

1. The origin of pairing
2. The self-consistent nature of the BCS
3. The gap equation
4. Analytical solution
5. The more general Hartree-Fock-Nambu formalism
6. The generalized gap equation for $\alpha = 0$
7. The full numerical solution

III. NANOWIRE MODELS

First and foremost, we showcase Lutchyn's minimal Majorana model [?] as our model's groundwork. Lutchyn's minimal model consists of a finite one-dimensional nanowire in the presence of an external Zeeman field V_Z and spin-orbit coupling (SOC) effect, being

described by the Hamiltonian

$$\hat{\boldsymbol{H}}_0 = \sum_{\sigma_i \sigma_j} c^\dagger_{\sigma_i} \left(-\eta \partial_x^2 - \mu + \alpha_{\mathrm{R}} i \sigma_y \partial_x + V_{\mathrm{Z}} \sigma_z \right) c_{\sigma_j}, \quad (1)$$

with $\eta = \hbar^2/2m^*$, m^* the electron's effective mass, and μ the chemical potential. The SOC effect due an externally applied transverse magnetic field of Rashba strength α_{R} and the Zeeman splitting $V_{\mathrm{Z}} = g_J \mu_B B_x/2$ due to a in-plane magnetic field B_x, with g_J the Landé $g$yromagnetic factor of the semiconductor and μ_B Bohr's magneton. Furthermore, Lutchyn then considers a s-type superconductive (SC) proximity effect of uniform pairing Δ_0 reading as $(\Delta_0 c^\dagger_\uparrow c^\dagger_\downarrow + h.c)/2$.

The major step up we make to this model is that we replace the *uniform* superconductive proximity effect term for a self-consistent onsite Hubbard self-energy term, which is analytical derived from a Hartree-Fock-Nambu mean-field formalism in appendix. Concretely, within the Nambu tight-binding scheme (also presented in that appendix), defining the spinor $\check{\boldsymbol{c}}_i = (c_{i\uparrow}\ c_{i\downarrow}\ c^\dagger_{i\uparrow}\ c^\dagger_{i\downarrow})$, the full Hamiltonian matrix reads explicitly as

$$\begin{aligned} \boldsymbol{H} = & \left(f\delta_{ij} - t\delta_{ij\pm1}\right)\left[\tau_z \otimes \sigma_0\right] + V_{\mathrm{Z}}\left[\tau_z \otimes \sigma_z\right]\delta_{ij} \\ & + \frac{\alpha_{\mathrm{R}}}{2a_0}\delta^{\pm}_{ij\pm1}\left[\tau_z \otimes i\sigma_y\right] + \boldsymbol{\Sigma}_{ij} \end{aligned} \quad (2)$$

where

$$\boldsymbol{\Sigma}_{ij} = \frac{1}{2}U\left[\mathrm{Tr}\left(\left[\tau_z \otimes \sigma_0\right]\tilde{\boldsymbol{\rho}}_{ii}\right)\left[\tau_z \otimes \sigma_0\right] - 2\tilde{\boldsymbol{\rho}}_{ij}\right]\delta_{ij} \quad (3)$$

with $f = 2t - \mu$ the band filling ($\mu = 0$ and $\mu = 2t$ corresponding to half and zero filling respectively), $t = \eta/a_0^2$ the hopping parameter. The Kronecker delta δ_{ij} denotes the on-site terms and $\delta^{\pm}_{ij\pm1}$ the 1st neighbors hoppings (front and back). Both σ and τ are Pauli matrices operating in their respectively spin and particle-hole subspaces, and

$$\tilde{\boldsymbol{\rho}}_{ij} = \begin{pmatrix} \boldsymbol{\rho}^{\mathrm{ee}}_{ij} & -\boldsymbol{\rho}^{\mathrm{eh}}_{ij} \\ -\left(\boldsymbol{\rho}^{\mathrm{eh}}_{ji}\right)^* & -\boldsymbol{\rho}^{\mathrm{ee}}_{ji} \end{pmatrix}$$

is a newly defined Bogoliubov–de Gennes (BdG) symmetric object relating to the Nambu reduced density matrix (rDM) $\boldsymbol{\rho}_{ij} = \langle \check{\boldsymbol{c}}^\dagger_j \otimes \check{\boldsymbol{c}}_i \rangle$ of the system with $\boldsymbol{\rho}^{\mathrm{ee}}_{ij} = \langle \boldsymbol{c}^\dagger_j \boldsymbol{c}_i \rangle$ and $\boldsymbol{\rho}^{\mathrm{eh}}_{ij} = \langle \boldsymbol{c}_j \boldsymbol{c}_i \rangle$. Their underlying spin structure ($\uparrow\uparrow$, $\uparrow\downarrow$; $\downarrow\uparrow$, $\downarrow\downarrow$) is being concealed within the bold notation. Therefore, in a straightforward and unpretentious manner, the significant improvement we have done to the Lutchyn model is that instead of the term $\Delta_0\left[\tau_y \otimes \sigma_y\right]\delta_{ij}$ we write instead the self-consistent onsite Hubbard self-energy term $\boldsymbol{\Sigma}_{ij}$, corresponding to the last line of Eq.(9).

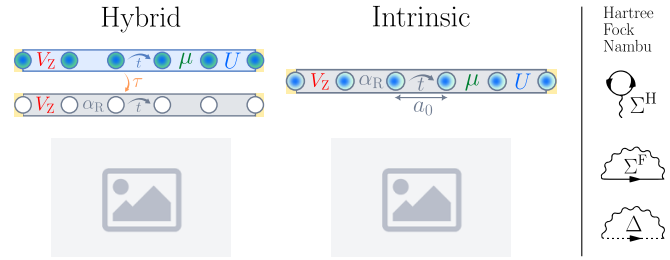

Figure 2. Models with DOS and self-energies diagrams

IV. RESULTS

Concerning with the sign of the on-site Hubbard potential U in a metal if on one hand $U < 0$ then the correlation $\rho^{\mathrm{eh}}_{i\uparrow j\downarrow}$ is favored and the system turns superconductive but if on the other hand $U > 0$ then the correlation $(\rho^{\mathrm{ee}}_{i\uparrow j\uparrow} - \rho^{\mathrm{ee}}_{i\downarrow j\downarrow})/2$ is instead favored and the system turns ferromagnetic. However, although $U < 0$ is a prerequisite for a non-zero superconductive correlation it alone does not guarantee it; this can be, for example, due to a strong enough Zeeman effect that forcefully magnetizes the system or due to a sufficient large μ since BCS theory tells us that $\Delta_{\mathrm{BCS}} \propto \exp[-Ua_0 n_{1\mathrm{D}}(\mu)]^{-1}$, with $n_{1\mathrm{D}}$ being the 1D density of states (DOS) decreasing as μ increases.

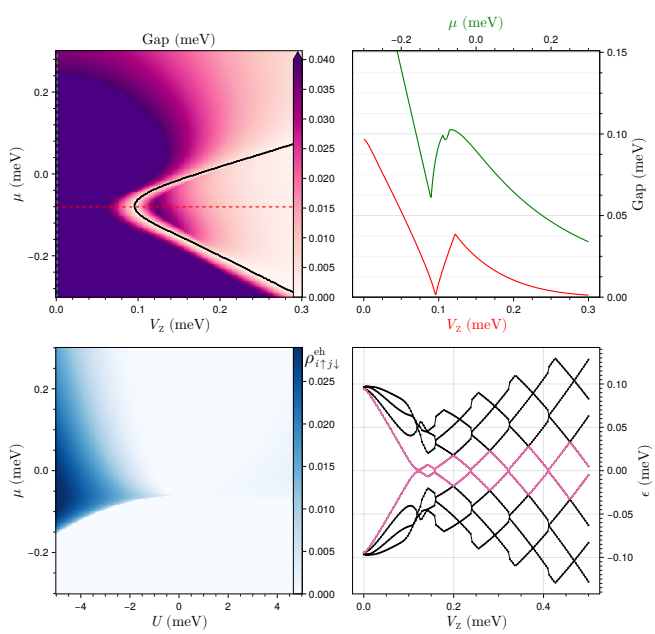

Figure 3. Hybrid results

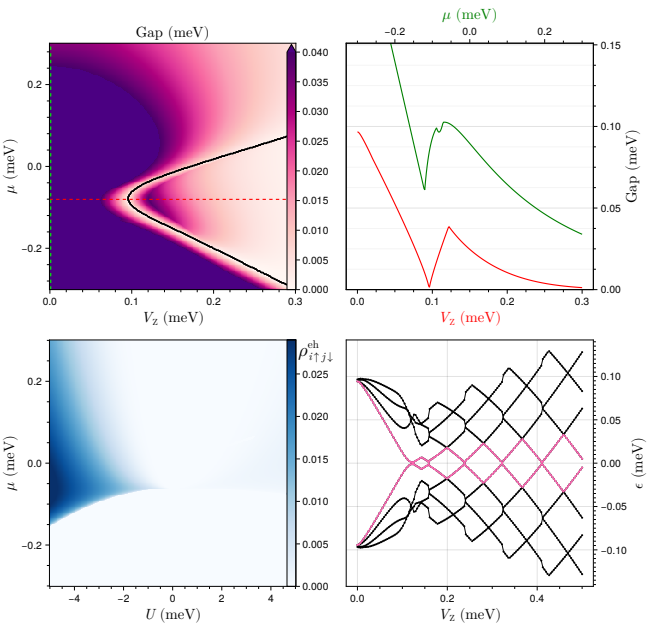

Figure 4. Intrinsic results

V. CONCLUSIONS

ACKNOWLEDGMENTS

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APPENDIX A: HARTREE-FOCK-NAMBU MEAN-FIELD THEORY

Consider a generic tight-binding quartic Hamiltonian,

$$\hat{H} = \sum_{\alpha\beta} h_{\alpha\beta} c^\dagger_\alpha c_\beta + \frac{1}{2}\sum_{\alpha\beta\gamma\delta} V^{\alpha\beta}_{\gamma\delta} c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta, \tag{4}$$

with $c^\dagger_\alpha(c_\alpha)$ fermionic creation (annihilation) operators of site and spin degree of freedom $\alpha = (i, \sigma_i)$, and $V^{\alpha\beta}_{\gamma\delta}$ a spinless electrostatic potential tensor of symmetries $V^{\alpha\beta}_{\gamma\delta} = V^{\beta\alpha}_{\delta\gamma}$ and $(V^{\alpha\beta}_{\gamma\delta})^* = V^{\gamma\delta}_{\alpha\beta}$ following directly from $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$. We fabricate a doubling of the degrees of freedom by introducing the Nambu-spinor $\check{c}_a = (c_a \;\; c^\dagger_a)$, constraining a particle-hole symmetry. In this Nambu representation the system's reduced density matrix (rDM) corresponds to the tensor product $\rho_{ab} = \langle \check{c}^\dagger_b \otimes \check{c}_a \rangle$, reading explicitly as

$$\rho_{ab} = \begin{pmatrix} \langle c^\dagger_b c_a\rangle & \langle c_b c_a\rangle \\ \langle c^\dagger_b c^\dagger_a\rangle & \langle c_b c^\dagger_a\rangle \end{pmatrix} = \begin{pmatrix} \rho^{\rm ee}_{ab} & \rho^{\rm eh}_{ab} \\ \rho^{\rm he}_{ab} & \rho^{\rm hh}_{ab} \end{pmatrix}, \tag{5}$$

with it's terms relating to each other directly from the fermionic anti-commutator properties.

Performing a mean-field Hartree-Fock decoupling as follows $\langle c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta\rangle \approx \rho^{\rm he}_{\beta\alpha}\rho^{\rm eh}_{\delta\gamma} - \rho^{\rm ee}_{\gamma\alpha}\rho^{\rm ee}_{\delta\beta} + \rho^{\rm ee}_{\delta\alpha}\rho^{\rm ee}_{\gamma\beta}$, the Nambu rDM equation of motion can be shown to read simply as $i\hbar\dot{\rho}_{ab} = \left[H^{\rm BdG}_{ab}, \rho_{ab}\right]$ with

$$H^{\rm BdG}_{ab} = \begin{pmatrix} h^{\rm HF}_{ab} & \Sigma^\Delta_{ab} \\ \left(\Sigma^\Delta_{ba}\right)^* & -h^{\rm HF}_{ba} \end{pmatrix} \tag{6}$$

the effective Bogoliubov-de Gennes (BdG) Hamiltonian, $h_{\rm HF}(\rho) = h + \mathbf{\Sigma}_H(\rho) + \mathbf{\Sigma}_F(\rho)$ the Hartree-Fock Hamiltonian, having defined the Hartree, Fock and pairing self-energies as

$$\begin{aligned} \Sigma^H_{ab} &= \textstyle\sum_{\gamma\delta} V^{\delta a}_{b\gamma}\rho^{\rm ee}_{\gamma\delta} \\ \Sigma^F_{ab} &= -\textstyle\sum_{\gamma\delta} V^{a\delta}_{b\gamma}\rho^{\rm ee}_{\gamma\delta} \end{aligned} \quad \text{and} \quad \Sigma^\Delta_{ab} = \sum_{\gamma\delta} V^{ab}_{\delta\gamma}\rho^{\rm eh}_{\gamma\delta} \tag{7}$$

Note the pairing symmetry $\Sigma^\Delta_{ab} = -\Sigma^\Delta_{ba}$.

Furthermore, if one considers the orbitals to ultra-localized $V^{\alpha\beta}_{\gamma\delta} \approx V(\mathbf{r}_\alpha - \mathbf{r}_\beta)\delta_{\alpha\delta}\delta_{\beta\gamma}$, the total self-energy of the system will amount to

$$\mathbf{\Sigma}_{ij} = \frac{1}{2}\sum_\ell V^{i\ell}\mathrm{Tr}\left(\left[\tau_z\otimes\sigma_0\right]\tilde{\boldsymbol{\rho}}_{\ell\ell}\right)\left[\tau_z\otimes\sigma_0\right]\delta_{ij} - V^{ij}\tilde{\boldsymbol{\rho}}_{ij},$$

with both σ and τ Pauli matrices operating in their respectively spin and particle-hole subspaces, and $\tilde{\boldsymbol{\rho}}_{ij}$ a newly defined BdG-symmetric object,

$$\tilde{\boldsymbol{\rho}}_{ij} = \begin{pmatrix} \boldsymbol{\rho}^{\rm ee}_{ij} & -\boldsymbol{\rho}^{\rm eh}_{ij} \\ -\left(\boldsymbol{\rho}^{\rm eh}_{ji}\right)^* & -\boldsymbol{\rho}^{\rm ee}_{ji} \end{pmatrix}. \tag{8}$$

Note the bold notation for the underlying spin structure ($\uparrow\uparrow$, $\uparrow\downarrow$; $\downarrow\uparrow$, $\downarrow\downarrow$). In this Wannier limit the Hartree self-energy stays exclusively onsite. Moreover, if one further considers the limiting case of a spin-1/2 Hubbard model $V^{ij} = U\delta_{ij}$, one obtains the purely onsite self-energy

$$\mathbf{\Sigma}_{ij} = \frac{1}{2}U\left[\mathrm{Tr}\left(\left[\tau_z\otimes\sigma_0\right]\tilde{\boldsymbol{\rho}}_{ii}\right)\left[\tau_z\otimes\sigma_0\right] - 2\tilde{\boldsymbol{\rho}}_{ij}\right]\delta_{ij}. \tag{9}$$

APPENDIX B: NUMERICAL DETAILS

All numerics done for this paper are available, well documented and primed for testing at the repository [???] written in *Julia* [???].

Concerning the optimization of the self-consistent routine $\rho_0 \to \Sigma_0 \to H_0 \to \rho_1 \to \Sigma_1 \to$ ..., since our Hubbard model contains exclusively onsite terms, one can encode all necessary information into the serializer $s = \left(h^{\rm HF}_{11}, h^{\rm HF}_{12}, h^{\rm HF}_{22}, \Delta_{12}\right)$, which in turn also guarantees that the BdG symmetries of the Hamiltonian are always maintained (as opposed to numerical errors progressively breaking it more and more after each iterations). One then decodes s to construct H and calculate the respective rDM. For the finite case we straightforwardly perform a full spectrum diagonalization [???], meaning that $\rho = -i(\epsilon + i\eta - H)^{-1}$, while for the infinite case we perform instead a *Schur's factorization* [???] where the rDM

is calculated through $\rho_{dn} = \frac{1}{2\pi}\int_{-\pi}^{+\pi} d\phi f(H(\phi) - \mu I)e^{idn\phi}$ with ϕ the Bloch phase and f the Fermi function at zero temperature. As for the fixed-point computations, we utilized the so called *Anderson acceleration/mixing* method [???].

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